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How policies impact household waste-source separation: an exploratory study of China

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ABSTRACT

This study explores the impact of policies on household separation behavior. A theoretical framework integrating the Technology Acceptance Model and the Theory of Planned Behavior was proposed to elucidate this relationship. The primary finding revealed that policy perception is the core explanatory variable linking policy to the behavioral changes. Furthermore, the mechanisms of policy perception were identified, showing that perceived ease of use and perceived usefulness represent two types of policy perceptions, each of which impacts behavior differently. Due to its exploratory nature, the findings warrant validation through further statistical analyses.

1. Introduction

The development of the global economy and improved living standards have driven overall consumption growth, resulting in rapid waste generation and disposal. Evidence shows that over three billion tons of municipal solid waste (MSW) are generated globally each year (Behera et al., 2020; Giraldo-Almario et al., 2024). Such large amounts of MSW are not only harmful to human health but also pose a serious threat to the ecological environment (Khatry et al., 2024; Llanquileo-Melgarejo and Molinos-Senante, 2022). Consequently, there is an urgent need for effective measures to achieve proper MSW management in the country.

The MSW management system mainly includes waste-source separation (separation), transport, processing and transformation, and disposal (Moloudi et al., 2022). Separation, as the most crucial step, determines the quantity and quality of waste flow in subsequent processing procedures and significantly impacts the comprehensive utilization of MSW (Ma et al., 2018). Therefore, promoting waste separation is essential for achieving effective MSW management.

However, because most people do not voluntarily engage in waste separation, governments must implement specific policies to intervene in such behavior (Hou et al., 2020; Pires et al., 2011). Except for a few industrialized countries (e.g., Japan), the deployment of separation policies remains limited in many nations (Ma et al., 2023b). Therefore, the effectiveness of separation policies has attracted the attention of researchers. Some studies have shown that policies may play a positive role. For example, Li et al. (2020), Liao et al. (2018), and Zhang et al. (2022) believed that separation policies could affect residents' intentions to separate waste. However, other studies have presented a more critical view. Issock et al. (2020) found that the separation policy had no effect on social and moral norms regarding separation intention. Pedersen and Manhice (2020) noted that the European Union's separation policy has not been as successful as expected. Han and Zhang (2018) pointed out that the separation policy in China has an insignificant influence on reducing the per-capita quantity of MSW.

Although the research mentioned above contributed to exploring the effectiveness of the separation policy, the analysis has three key limitations. First, most studies lack rigorous experimental designs, such as randomized controlled trials, making it challenging to isolate policy effects from confounding variables, including rising education levels, which also influence separation behavior (Miltojevic et al., 2017). Second, while some studies acknowledge a "black box" between policy and behavior (Yin et al., 2022), few treat the policy itself as a core explanatory variable, leaving the underlying mechanisms of influence under-explored. Third, prior studies have focused primarily on economic and environmental outcomes (Lavee, 2007; Matsumoto, 2011), with limited attention paid to the behavioral effects. As behaviorally

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informed policy research gains traction (Bhanot and Linos, 2020; Ewert et al., 2021), understanding how policies influence individual behavior is crucial for enhancing their effectiveness. However, current environmental policy research often lacks this behavioral perspective, limiting its practical relevance (Knickmeyer, 2020; Wu et al., 2021).

In summary, new problems have emerged in the field. Does the policy play a role in separation behavior, and how can its effectiveness be ensured? These questions are critical because the pursuit of policy effectiveness is the eternal goal of the government. The research findings will assist governments in implementing targeted measures to enhance MSW management, address the gap in the behavioral perspective of environmental policy research, and gain a better understanding of how policy effects are generated.

To address existing research gaps, this study treats policy perception as a core explanatory variable to uncover the mechanism linking policy implementation and household separation behavior. As external policies influence behavior only through individual perception (Kurt, 2017), we adopted a quasi-experimental design to examine this relationship and to explore its underlying pathways. Given China's status as the world's largest generator of municipal solid waste (Zhang et al., 2023) and the issuance of over 180 separation-related policies since 1986 (Fan et al., 2023), we selected a Chinese city as our study site. This context not only highlights the urgent need for effective waste management but also provides valuable insights applicable to other developing countries. This study further contributes to the literature by collecting first-hand field data through participant interviews, offering empirical evidence for evaluating policy effectiveness from a behavioral perspective.

As an exploratory study, our primary aim is to uncover initial patterns and relationships rather than to confirm hypotheses with high statistical certainty. Although the sample size is relatively large, we acknowledge that a more rigorous statistical analysis would be required for confirmatory research.

2. Literature review and theoretical framework

2.1. The role of policy perception

Scholars have long realized that the primary goal of public policy is to influence or shape people's behavior (Schneider and Ingram, 1990). Thus, understanding the relationship between policy and behavior has become a central concern in policy research in recent years. As an individual's perception of policy-related information, policy perception has been considered to influence people's behaviors, including green entrepreneurship (Chu et al., 2021), technology adoption (Chen and Chen, 2017), and political participation (Hellwig et al., 2010).

The widespread implementation of waste separation policies has prompted researchers to examine the behavioral effects of policy perception. For example, Negash et al. (2021) studied this relationship in Mongolia, Shen et al. (2022) in urban and Liao et al. (2018) in rural China. However, these findings are inconsistent. While Wan et al. (2014) argued that policy perception directly influences separation behavior, Xu et al. (2017) found no significant relationship between the two.

These inconsistencies may have stemmed from methodological limitations. Most existing studies did not adopt quasi-experimental designs and often lacked proper allocation concealment and outcome masking. Critically, many studies have assumed that all participants were exposed to and influenced by policy perception, overlooking the fact that individuals unaware of the policy should not be included in analyses of its effects on perceptions. According to Lewin's Psychological Field Theory, policy perception is the primary channel through which policies influence behavior (Lewin, 2017). Rigorous empirical methods are therefore needed to determine whether policy perception genuinely affects separation behavior, which, in turn, reflects the effectiveness of the policy itself.

Moreover, prior research rarely explores the multidimensional nature of policy perceptions. Most studies treat it as a unidimensional construct, for instance, using perceived policy effectiveness as a proxy (e.g., Wan et al., 2014). This approach neglects the internal complexity of policy perception, despite growing evidence that it comprises multiple dimensions (Pierce, 2014; Sihombing and Permana, 2023).

2.2. Driving factors of household waste separation

Waste separation offers substantial benefits for resource conservation and environmental protection. However, efforts to promote waste separation have yielded limited success in many countries, particularly in developing nations (Ma et al., 2023b). Some studies have examined the factors that motivate households to separate their waste. For example, Xu et al. (2018) emphasized the role of social-psychological factors, and Ma et al. (2018) highlighted the importance of situational influences. Rousta et al. (2020) identified attitude, moral norms, subjective norms, and perceived behavioral control as the most influential factors in developing countries, as determined through a meta-analysis. Interestingly, despite extensive research, sociodemographic variables were found to have the weakest impact on the results.

Other studies have explored additional behavioral factors. Sunarti et al. (2021) found that knowledge and emotional factors significantly enhance participation in separation activities. Chen et al. (2022) demonstrated that information intervention can effectively promote separation. Yin and Ma (2022), in a survey conducted in Xi'an, China, revealed that altruistic motivations also play a role.

While previous research has provided a relatively comprehensive overview of the factors influencing waste separation, few studies have considered policy perception as a key determinant of waste separation behavior. This gap limits our understanding of how individuals' interpretations of policy initiatives shape their behavioral responses.

2.3. The application of the Theory of Planned Behavior

Separation behavior has been widely examined through social-psychological analysis frameworks such as the Theory of Planned Behavior (TPB) (Ajzen, 1991), Value-Belief-Norm Theory, and Attitude-Behavior-Condition Theory. Among these, the TPB is the most extensively applied theory. Developed from the Theory of Reasoned Action, the TPB posits that human behavior is not entirely voluntary but is shaped by multiple psychological factors.

The standard TPB consists of five elements: attitude, subjective norms, perceived behavioral control (PBC), behavioral intention, and behavior. *Attitude* reflects individuals' evaluations of the likely outcomes of a behavior, *subjective* norm captures perceived social pressure to perform or not perform the behavior, and *PBC* refers to individuals' perceived ease or difficulty in performing the behavior. These three factors indirectly or directly influence behavior by shaping behavioral intentions, particularly in the case of PBC. *Behavioral intention* refers to an individual's motivation or willingness to engage in a specific behavior, whereas *behavior* refers to the actual action performed by an individual.

The TPB has demonstrated strong explanatory power in various contexts. For example, Ma et al. (2018) found that attitudes and subjective norms significantly influenced behavioral intentions toward waste separation. Similarly, Ari and Yilmaz (2016) and Chu and Chiu (2003) identified PBC as the most critical determinant of behavioral intention. The robustness of the TPB has been further supported by recent meta-analyses (Ma et al., 2023a).

A key strength of the TPB is its flexibility, which allows for the integration of additional variables to enhance its explanatory power (Ajzen, 1991). Consequently, many studies have extended the TPB framework by incorporating constructs such as awareness of consequences (Tonglet et al., 2004), moral norms (Chu and Chiu, 2003), and self-identity (Pakpour et al., 2014), often yielding improved model performance (Ma et al., 2018). Therefore, we believe that the TPB can serve as a theoretical foundation for analyzing the mechanism of policy perception about separation behavior.

2.4. Theoretical framework

Recent studies have shown that the constructs of the Technology Acceptance Model (TAM), that is, perceived usefulness (PU) and perceived ease of use (PEOU), are applicable not only in the context of technology usage but also in the application of the TAM within policy contexts. Notably, Pierce (2014) extended the TAM to propose the Policy Acceptance Model (PAM), in which PU and PEOU are fully integrated as dimensions of policy perception. Additionally, PU and PEOU have been applied in specific policy contexts, such as air pollution reduction (Rotaris and Danielis, 2014) and license plate restriction policies (Jia et al., 2017). Based on these studies and the intrinsic link between the TAM and the separation policy, we believe that PU and PEOU can be effectively applied to the implementation of the separation policy.

Specifically, the TAM is used to examine how external variables influence an individual's adoption and use of a particular system (Davis, 1989). The separation policy primarily provides technical services (e.g., separation knowledge and skills), equipment support (e.g., separation bins), and guidance to residents. The effectiveness of the separation policy depends on perceptions about and use of this separation system, which subsequently translates into separation behaviors. Therefore, the TAM constructs were considered in this study.

Furthermore, both the TAM and the TPB are derived from the Theory of Reasoned Action (Ajzen, 1991; Davis, 1989). While PU and PEOU primarily focus on the impact of external factors on behavior, the TPB emphasizes the role of individual and internal factors in shaping behavior. Therefore, integrating the TAM and the TPB (i.e., expanding the TPB with PU and PEOU) provides a more comprehensive framework for incorporating psychological latent variables, enabling a deeper investigation of the impact of external factors on behavior (Smith and Brown, 2023; Yan et al., 2023). Additionally, waste separation behavior has been recognized as rational behavior in previous research (Yin and Ma, 2022). Therefore, this study integrates the constructs of PU and PEOU into the TPB to examine the impact of policy as an external factor on household waste separation.

PU of a policy is defined as the degree to which a person believes that the policy measures would address the policy issue (Xu et al., 2022), which stems from the perception of the overall effectiveness of the policy. According to Self-Efficacy Theory, if residents believe that the separation policy is effective in addressing the issue of "garbage siege" and enhancing their living environment, it fosters positive outcome expectations, also known as positive perceived usefulness (PU). This belief instills a high sense of self-efficacy in individuals, thereby encouraging them to comply with separation policy requirements and potentially adopt separation behaviors (Basileo et al., 2024). Meanwhile, based on the trust relationship with the government, residents with high PU may be more willing to achieve the government's goals (Yin et al., 2022). Therefore, PU may significantly affect household waste separation. Thus, the following hypotheses are proposed.

H1. PU has a positive influence on separation intention.

H2. PU has a positive influence on separation behavior.

PEOU of a policy refers to the degree to which a person believes that using a policy support system would be effortless (Sihombing and Permana, 2023). Separation policies usually provide a convenient support system for households to facilitate waste separation. In addition to clear guidelines and separation bins, the system includes skill training and knowledge building. Thus, it is reasonable to assume that if residents perceive the support system as convenient, they will develop a sense of PEOU. According to Nudge Theory, the convenient support system design provided by the separation policy is entirely based on the needs of behavioral performance, aiming to align as closely as possible with people's habits and preferences. This approach enhances individuals' PEOU, thereby facilitating the achievement of behavioral intervention goals (Amiri et al., 2024). Effort can also be viewed as a finite resource that people allocate to various activities (Hamid et al., 2016). Naturally, residents are more likely to take actions that they perceive to be easier to perform than others. Therefore, we propose the following hypotheses.

H3. PEOU positively affects separation intention.

H4. PEOU has a positive effect on separation behavior.

Furthermore, both the TAM and the PAM indicate that PEOU can influence PU (Pierce, 2014). According to the Cost-Benefit Theory, if the effort required to comply with a policy exceeds the benefits that can be obtained, people may not adhere to the policy. Thus, the less effort needed to comply with a policy, the greater the relative benefits. Therefore, we assume that.

H5. PEOU positively impacts PU.

Psychologists argue that stimuli and behavioral responses are mediated by internal psychological processes that, although not directly observable, can be inferred from their antecedents and outcomes (Keulen et al., 2022; Mills, 2002). In response to policy stimulus, policy perception may influence behavior through mediating processes. Based on the TPB, an individual's behavioral intention is influenced by a combination of attitude, subjective norms, and PBC. In the context of waste separation, attitudes reflect residents' relatively stable evaluations of their separation behavior. The more they understand its benefits, the more favorable their attitudes are. According to the concept definition, PU can help residents understand the potential benefits of separation, leading them to develop a positive attitude toward it. This attitude, in turn, encourages them to make separation decisions. PBC of separation refers to whether households can bear the additional costs of adopting separation behavior, which is influenced by conditional assessment and perception ability. PEOU represents the degree of alignment between individual abilities and the knowledge and skills required to use a new system (Mathieson, 1991). The higher the PEOU, the greater the residents' grasp of separation knowledge and skills, forming a higher level of PBC, which in turn affects household waste separation. Thus, household separation behavior tends to be a conscious pursuit of goals (Yin and Ma, 2022), while the TPB emphasizes the role of conscious choice in the process of goal pursuit (Xie and Liu, 2013). Many studies have also confirmed that attitude and PBC play a mediating role between policy perception and individual behavior (Yin et al., 2022; Wang et al., 2021). Policy perception may influence individual attitudes and mitigate behavioral challenges in policy compliance. Therefore, we considered that PEOU and PU may indirectly affect the separation behavior, according to the TPB. Thus, the following hypotheses are proposed.

H6. PU has a positive influence on attitudes toward separation.

H7. PEOU has a positive influence on the PBC of separation.

At the same time, the premise for supporting policy perception to influence separation behavior indirectly is that the TPB can adequately explain separation behavior. Attitude comprises three components: cognition, emotion, and behavioral tendencies. The cognitive component involves an individual's beliefs and opinions about a particular behavior, the emotional component consists of an individual's emotional response to that behavior, and the behavioral tendency component consists of an individual's readiness to engage in that behavior. This tripartite structure enables attitudes to comprehensively influence an individual's behavioral intention (NeuroLaunch Editorial Team, 2024). PBC involves evaluating an individual's resources and opportunities. When individuals believe that they have sufficient resources and opportunities to perform a particular behavior, their behavioral intentions become stronger (Ajzen, 1991). Behavioral intention reflects an individual's motivation and determination to engage in a specific behavior. The stronger the motivation, the more likely an individual is to engage in actual behavior. Thus, the following additional hypotheses are proposed.

H8. Attitude is positively related to separation intention.

H9. PBC is positively related to separation intention.

H10. Separation intention is positively related to separation behavior.

3. Analytic strategy

This study was divided into two sections. Before verifying the theoretical framework, we examined whether policy perception could impact separation behavior, as it is difficult to infer from existing research evidence with accuracy. Therefore, the primary aim of the first analysis was to examine whether policy perceptions affect separation behavior. Given the significant impact of policy perception on separation behavior, as in the second analysis, we verified the proposed theoretical framework to reveal the "black box" mentioned in the Introduction. A key premise supporting the proposed theoretical framework is that the separation policy generates policy perceptions, which further influence separation behaviors. In the second analysis, we analyzed the relationship between policy perception and household waste separation and found that policy perception is the primary mechanism through which policy influences household behavior.

4. Analysis 1

In the first analysis, a quasi-experimental design was employed, following Angrist and Imbens (Angrist et al., 2017; Imbens, 2010), who advanced the idea that quasi-experiments can reliably explore the causal relationship between variables.

4.1. Method

4.1.1. Quasi-experimental research design

This study aimed to examine whether policy perception causally influences household waste separation. A direct intervention study is not feasible because the separation policy has already been implemented and functions as an exogenous shock to this system. Moreover, residents' policy perception is not subject to self-selection (Chen and Lee, 2020), and researchers cannot manipulate or intervene in individuals' perceptions (Creswell and Creswell, 2018). To address this, we adopted a quasi-experimental design that lies between observational studies and randomized controlled trials. This approach uses naturally occurring exogenous shocks and observational data to simulate the experimental conditions (Liu et al., 2021). Individuals exposed to the policy were assigned to the experimental group, while those who were not were assigned to the control group. To improve comparability, a sample trimming technique was applied to align the two groups with the assumptions of a randomized controlled trial (Ma et al., 2021).

In this study, intervention is defined as policy perception, which refers to residents' awareness and understanding of the separation policy and related activities. Although the policy was implemented uniformly across the study area, variations in information dissemination and individual receptiveness resulted in differences in residents' exposure to policy-related information across the three counties. Consequently, some residents developed a clear perception of the policy (experimental group), while others did not (control group), despite living in the same administrative region. We conducted brief oral interviews to evaluate residents' perceptions and understanding of the separation policy. This qualitative approach allowed us to distinguish between those who had been meaningfully exposed to the policy and those who had not.

Given the potential for demographic differences between the two groups, we employed a sample trimming method to enhance comparability (Ma et al., 2021; Yin et al., 2022). Specifically, we used Excel to generate random numbers and randomly removed excess individuals from the control group to match the demographic structure of the intervention group. Trimming was based on key demographic variables, including gender, age, income, and education.

We conducted chi-square tests on the control variables to verify the effectiveness of these adjustments. The results showed no statistically significant differences between the two groups after trimming, indicating that the groups were comparable in all major demographic aspects except for the presence or absence of policy perception.

The quasi-experiment was implemented through a field survey using two questionnaires: Questionnaire A for the experimental group and Questionnaire B for the control group. The key difference is that Questionnaire B excludes items that measure policy perception. Both questionnaires assessed separation behavior, demographic characteristics, and other potential determinants, such as attitude, perceived behavioral control (PBC), and separation intention. All items were measured using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree) (see Fig. 1).

4.1.2. Site selection

As shown in Fig. 2, Guilin, located in the northeast of the Guangxi Zhuang Autonomous Region of China, was selected as the study area for this research. Guilin has been one of the eight pilot cities that promoted separation in China. Unlike other pilot cities (i.e., Beijing, Shanghai, Guangzhou, Shenzhen, Hangzhou, Nanjing, and Xiamen), Guilin is more representative of Chinese cities because it is located inland and has a GDP that is approximately the national average (Ma et al., 2018).

4.1.3. Participants

In August 2022, a field survey was conducted in the central urban areas of Guilin, specifically the Xiufeng, Diecai, Xiangshan, Qixing, and Lingui districts. The survey adopted a stratified random sampling method to select participants. Specifically, using web crawler technology (according to online rental prices), two communities with high and low economic levels were randomly selected from each district in the city. Thus, the residents of the ten selected communities were the participants of this study. Over 1000 households participated in the survey, and 862 valid questionnaires were collected (Table 1).

The survey sample exhibited strong demographic alignment with national profiles. Adults aged 65 and above accounted for 19.03 % of respondents (versus China's 14.2 % national average; NBS, 2022), while children (7–14 years) represented 12.53 %, and the other cohorts showed balanced distributions (13–14 % each). This proportional representation accounts for the exclusion of children aged <7 years. Regarding income distribution, 46.98 % of the participants earned less than 2500 RMB per month, closely mirroring the national urban population pattern, where 40 % earn below 2300 RMB per month (NBS, 2021). These consistent patterns across both demographic and socioeconomic dimensions confirm the sample's robust national representativeness. At the time of the survey, Guilin's population was 4.956 million. Based on professional statistical calculations (e.g., using a Sample Size Calculator), a sample size of 862 yielded a sampling error of only 3.41 % at a 95 % confidence level and a 50 % response distribution (considered the most conservative scenario). Furthermore, this study employed a stratified sampling method to ensure proportional representation across key demographic groups. These factors collectively support the representativeness and scientific rigor of our sample.

In this study, the essential premise of the quasi-experiment was that the selection of subjects for the experimental and control groups was random (Ballance, 2023; Pellegrin et al., 2018). After grouping, it was found that the control group had more participants with lower education and younger age than the intervention group. To address this issue, this study used a sample-trimming method (Ma et al., 2021) to process the data from the experimental and control groups. As shown in Table 2, the chi-square test results indicated no significant differences in the demographic characteristics between the experimental and control groups ($p > 0.1$). Therefore, the participants in the two groups were effectively randomized, allowing for a randomized controlled analysis using a quasi-experimental design.

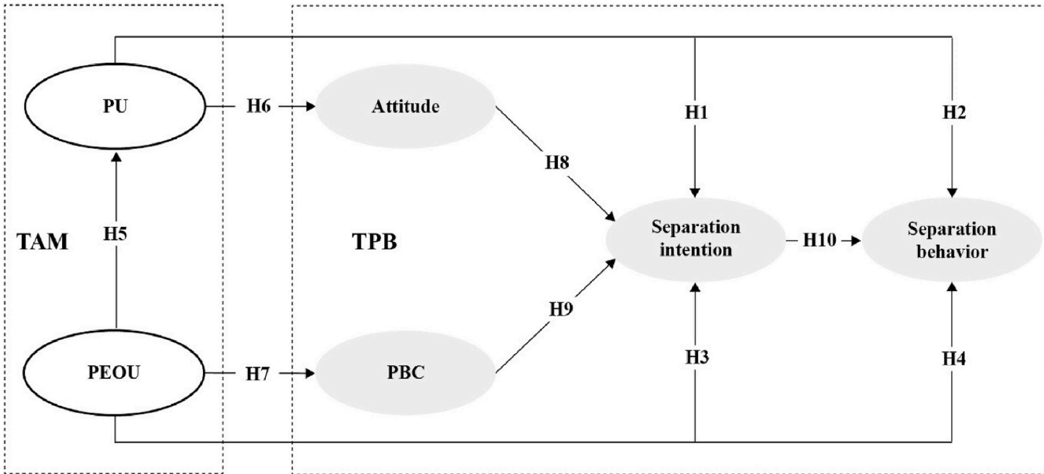

Fig. 1. Theoretical framework and hypothesis development.

Fig. 2. Guilin, China (Ma et al., 2018).

4.2. *Results*

As behavioral intention is an antecedent explanatory variable of behavior, separation intention and behavior were used as observational variables for household waste separation. As shown in Fig. 3, the mean separation intention with policy perception (Mean = 4.459, S.D. = 0.743) was significantly higher than that without policy perception (Mean = 4.095, S.D. = 0.920). Similarly, the mean separation behavior with policy perception (Mean = 3.505, S.D. = 1.056) was significantly higher than that without policy perception (Mean = 2.903, S.D. = 1.077). The results passed the homogeneity of variance test and were verified using an independent-sample *t*-test (separation intention: $p < 0.001$, $z = -4.754$; separation behavior: $p < 0.001$, $z = -6.314$). This finding suggests that accepting policy perception can enhance household waste separation. In other words, from the perspective of policy perception, the separation policy is effective.

5. Analysis 2

Building on the first analysis, the second study investigated the specific relationship between policy perception and separation behavior to verify the proposed theoretical framework. As found in the first analysis, not all residents demonstrate policy perception, and residents without policy perception cannot be used as research objects to analyze

Table 1
Descriptive statistics of the survey samples.

Category	Response	Frequency	Percentage
Gender			
	Male	335	38.86 %
	Female	527	61.14 %
Age			
	7–14	108	12.53 %
	15–24	112	12.99 %
	25–34	118	13.69 %
	35–44	122	14.15 %
	45–54	121	14.04 %
	55–64	117	13.57 %
	65 and above	164	19.03 %
Education level			
	Illiterate	17	1.97 %
	Elementary school	121	14.04 %
	Junior high school	190	22.04 %
	Senior high school	203	23.55 %
	Junior college or Bachelor's	260	30.16 %
	Master's and above	71	8.24 %
Monthly income			
	Below 2500RMB	405	46.98 %
	2501-4000RMB	198	22.97 %
	4001-6000RMB	132	15.31 %
	6001-8000RMB	68	7.89 %
	8001-10000RMB	32	3.71 %
	10001-20000RMB	19	2.20 %
	Over 20000RMB	8	0.93 %

Note: All participants included in the survey had lived in their communities for more than one year.

the mechanisms of policy perception.

5.1. Method

5.1.1. Structural equation modeling

To verify the theoretical framework, structural equation modeling (SEM) was employed using AMOS 23.0 to build a measurement model (Fornell and Larcker, 1981). SEM can integrate factor and path analysis methods, effectively deal with the structural relationships among multiple variables, and overcome collinearity among independent variables. The experimental group, which was affected by policy perception, served as the research sample.

5.1.2. Measures

Policy perception. According to previous studies (Pierce, 2014; Yin et al., 2022), measuring policy perception involves assessing an individual's favorable or unfavorable perception of a separation policy. PU was measured using six items, such as "the separation policy has improved residents' quality of life", "the separation policy is scientifically based and reasonable", and "the implementation effect of the separation policy is good". PEOU was measured using eight items, such as "the waste separation bins provided by the government are sufficient", "the waste separation bins provided by the government are conveniently located", and "the waste separation slogans provided by the government are easy to understand". The specific measurement contents are listed in Table 3.

Other measures. To measure other variables involved in this study, items used in previous similar studies were adopted (Ma et al., 2023b; Tonglet et al., 2004): (i) attitude was measured using four items (e.g., "waste separation is good"); (ii) five items were included to capture PBC (e.g., "whether or not I separate waste is entirely up to me"); (iii) separation intention was assessed using three items (e.g., "I plan to separate waste"); and (iv) separation behavior was measured using the item "I never/often separate my household waste."

5.1.3. Participants

The residents of the experimental group (n = 653) in the first analysis

Table 2
Distribution of demographic characteristics.

Category		Participants			chi-square
		Experimental group (n = 653)	Control group (n = 154)	Total (n = 807)	
Gender					0.001
	Male	251	59	310	
	Female	402	95	497	
Age					8.281
	Below 14	75	14	89	
	15–24	65	25	90	
	25–34	79	25	104	
	35–44	103	19	122	
	45–54	100	21	121	
	55–64	97	20	117	
	65 and above	134	30	164	
Education level					8.764
	Illiterate	12	5	17	
	Elementary school	77	23	100	
	Junior high school	162	28	190	
	Senior high school	172	31	203	
	Junior college or Bachelor's	191	55	246	
	Master's and above	39	12	51	
Monthly income					4.090
	Below 2500RMB	299	62	361	
	2501-4000RMB	149	42	191	
	4001-6000RMB	109	21	130	
	6001-8000RMB	50	16	66	
	8001-10000RMB	25	7	32	
	Over 10000RMB	21	6	27	

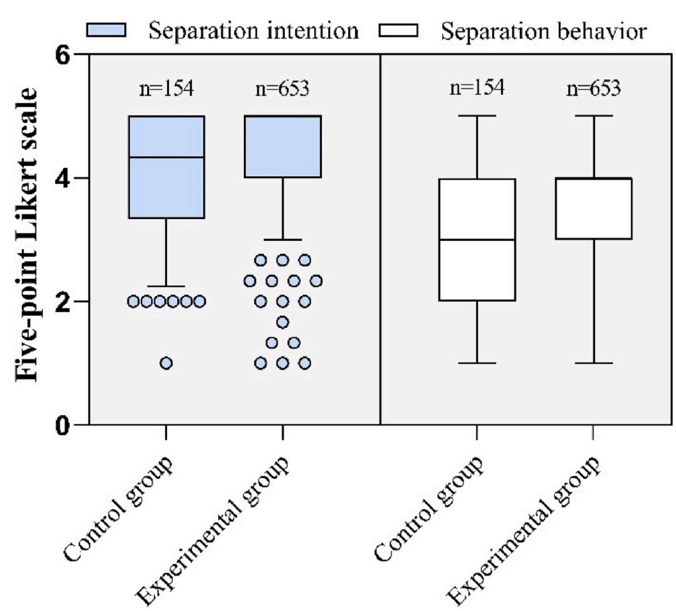

Fig. 3. Box plots of raw data of separation intention and behavior of the experimental and control groups.

Table 3
Reliability measurements for each item.

Constructs	Items	Item contents	Cronbach's alpha	AVE	Factor loadings	CR
Perceived ease of use (PEOU)	PEOU1	The waste separation bins provided by the government are sufficient.	0.898	0.525	0.626	0.898
	PEOU2	The waste separation bins provided by the government are conveniently located.			0.651	
	PEOU3	The waste separation signs provided by the government are decent.			0.701	
	PEOU4	The waste separation guidance provided by the government is clear.			0.825	
	PEOU5	The waste separation slogans provided by the government are easy to understand.			0.783	
	PEOU6	The government provides useful separation training activities on knowledge and skills.			0.807	
	PEOU7	The separation policy can help residents conveniently dispose of recyclables.			0.695	
	PEOU8	The separation policy enables me to separate waste.			0.683	
Perceived usefulness (PU)	PU1	The separation policy has improved the level of environmental management.	0.853	0.498	0.700	0.855
	PU2	The separation policy has improved residents' quality of life.			0.650	
	PU3	The separation policy is scientifically based and reasonable.			0.563	
	PU4	The implementation effect of the separation policy is good.			0.702	
	PU5	The separation policy is effective in reducing waste.			0.760	
	PU6	I am satisfied with the separation policy.			0.831	
Attitude (ATT)	ATT1	Waste separation is good.	0.778	0.552	0.715	0.786
	ATT2	Waste separation is rewarding.			0.831	
	ATT3	Waste separation is responsible.			0.674	
Perceived behavioral control (PBC)	PBC1	Waste separation is easy.	0.800	0.598	0.632	0.814
	PBC2	I have enough time to separate waste.			0.880	
	PBC3	Whether or not I separate waste is entirely up to me.			0.787	
Separation intention (SI)	SI1	I am willing to separate waste.	0.900	0.745	0.827	0.897
	SI2	I plan to separate waste.			0.908	
	SI3	I intend to separate waste.			0.852	

were selected as the research subjects for the second study. These residents have a clear perception of the policy. After excluding irrelevant research subjects, the identified mechanisms of policy perception on separation behavior will be more objective and accurate.

5.1.4. Follow-up interview

To better understand and verify the conclusions of this study, we conducted telephone interviews after the field surveys. Specifically, an interview outline was first designed based on the questions of interest (e.g., why do you think the separation policy is effective or ineffective?). Participants who provided their contact information were randomly selected and contacted. Telephone interviews were conducted with participants who gave their consent to participate in the study. The interviews were conducted according to the following guidelines: First, we introduced the background and purpose of the interview to the participants. Subsequently, according to the interview outline, we conducted semi-structured interviews with each participant. The conversations were recorded and subsequently converted to text. Content analysis of the original text data (Braun and Clarke, 2006) was then conducted, and the main information was summarized through coding and classification.

5.2. Results

5.2.1. The results of the measurement model

Before testing the hypotheses, a measurement model was developed using structural equation modeling. To assess its adequacy, a confirmatory factor analysis was conducted, which is appropriate for evaluating model reliability (Cosa et al., 2024). The analysis focused on the experimental group exposed to policy perceptions (n = 653). Using IBM SPSS AMOS 23, the model fit indices were as follows: (i) χ^2/df = 3.767 (<5), (ii) RMSEA = 0.065 (<0.08), (iii) RMSR = 0.044 (<0.10), (iv) CFI = 0.920 (≥0.90), (v) IFI = 0.920 (≥0.90), and (vi) TLI = 0.908 (≥0.90). As shown in Table 3, Cronbach's alpha values ranged from 0.778 to 0.900 (threshold >0.5), the AVE ranged from 0.498 to 0.745 (threshold ≈ 0.5), and the composite reliability ranged from 0.786 to 0.898 (threshold >0.6). All indices met the recommended thresholds (Kline, 2005; Purnomo, 2017), indicating that the measurement model was both reliable and valid.

As shown in Table 4, each construct's component score exceeds its correlations with other constructs, indicating good discriminant validity (Fornell and Larcker, 1981). To assess common method bias, a measurement model containing an unmeasured latent method factor was examined (Table 5). The single-factor model generated a poor fit compared to the measurement model. In contrast, the differences in chi-squared, RMSEA, and CFI between the measurement model with a latent method factor and the restricted model (controlling for the unmeasured latent method factor) were minimal (Podsakoff et al., 2012).

5.2.2. Hypothesis testing

After the model validation was completed, the path analysis results were obtained. To assess whether the sample size used in this study was sufficient to detect the hypothesized effects, we conducted a post-hoc statistical power analysis using the inverse square root method proposed by Kock and Hadaya (2018). This method estimates the minimum required sample size based on the smallest statistically significant path coefficient in the model. The formula is as follows.

$$\widehat{N} > \left(\frac{z_{0.95} + z_{0.80}}{|\beta|_{min}}\right)^2 \tag{1}$$

where $\widehat{N}$ represents the minimum required sample size, $|\beta|_{min}$ denotes the absolute value of the smallest statistically significant path coefficient, $z_{0.95}$ is the z-score corresponding to a 0.05 significance level (≈1.645), and $z_{0.80}$ is the z-score corresponding to 80 % statistical power (≈0.842).

In this study, the smallest statistically significant path coefficient was

Table 4
Test results for discriminant validity.

	PEOU	PU	PBC	ATT	SI
PEOU	**0.725**				
PU	0.632	**0.706**			
PBC	0.488	0.367	**0.773**		
ATT	0.279	0.304	0.149	**0.743**	
SI	0.437	0.376	0.621	0.360	**0.863**

Table 5
Results of multi-group confirmatory factor analysis.

Model	χ^2/df	RMSEA	CFI	Δ χ^2/df	Δ RMSEA	Δ CFI
Single-factor model	14.878	0.146	0.582	+11.111	+0.081	−0.338
The measurement model	3.767	0.065	0.920	–	–	–
The measurement model controlling for the unmeasured latent method factor	3.658	0.064	0.927	−0.109	−0.001	+0.007

0.147, resulting in the following:

$$\widehat{N} > \left(\frac{2.486}{0.147}\right)^2 \approx 287 \tag{2}$$

The actual sample size for SEM was 653, which exceeded the threshold, meaning that the study had sufficient statistical power to detect the effects of interest, supporting the robustness of our empirical findings.

As shown in Fig. 4, the results indicate that PU did not affect separation intention ($\beta = 0.052, p > 0.1$) or separation behavior ($\beta = -0.014$, $p > 0.1$), suggesting that H1 and H2 were not supported. PEOU did not affect separation intention ($\beta = 0.059, p > 0.1$), but it had a positive and significant effect on separation behavior ($\beta = 0.147, p < 0.05$). H3 was not supported, whereas H4 was supported. PU positively influenced attitude ($\beta = 0.253$, $p < 0.01$), PEOU positively influenced PBC ($\beta = 0.488$, $p < 0.01$), and the TPB effectively explained classification behavior (ATT→SI: $\beta = 0.278, p < 0.01$; PBC→SI: $\beta = 0.554, p < 0.01$; SI→SB: $\beta = 0.288, p < 0.01$). These findings support H6, H7, H8, H9, and H10, as well as the mediating role of PU and PEOU. In addition, PEOU had a positive and significant effect on PU. Thus, H5 is supported.

5.2.3. Qualitative exploration

To further contextualize the quantitative findings, we conducted a targeted qualitative analysis of some residents in the second analysis. This supplementary exploration aimed to elucidate participants' perspectives on policy awareness, thereby further clarifying the pathways and effects of policy perception.

A total of 80 residents participated in follow-up interviews. Based on the key information gathered from the participants' conversations, we identified the following key findings: (i) Opinions on PEOU. Many residents believed that the separation policy should provide convenience because waste separation is difficult. Difficulty in separating and distinguishing the categories of waste is the main reason why residents are unwilling to separate; (ii) Opinions on PU. Most residents considered that the separation policy lacked effectiveness, which is an important reason why they do not need to follow the policy requirements; (iii) Opinions on information intervention. The government has made great efforts in information intervention, but it seems to have no impact on separation behavior; and (iv) Opinions on the government. The government should design a more effective waste separation policy to enhance public participation.

6. Discussion

6.1. The mechanism of policy perception on separation behavior

By integrating the results of Studies 1 and 2, it can be inferred that policy perception positively impacts separation behavior. This influencing mechanism mainly includes the following.

i. **In this study, policy perception had a partial direct impact on separation behavior.** PU cannot directly influence either the intention or the behavior of separation. This finding is consistent with those of previous studies, such as Chen and Gou (2024). A reasonable explanation is that the current implementation of the separation policy does not provide residents with a direct sense of policy effectiveness because waste management is a long-term process (Liu, 2018). As a result, PU has not reached the "sensory threshold" required to drive behavioral changes (Huang and Nong, 2011). The interviews revealed that most residents believed the separation policy, which has been in place for many years, is ineffective, and therefore, there is no need to adhere to its requirements. PEOU cannot directly influence separation intention, but can directly influence separation behavior. Some researchers have pointed out that PEOU has no direct impact on behavioral intention (Aris et al., 2022), which is consistent with the findings of this study. Although behavioral intention is an explanatory variable for actual behavior (Meng et al., 2024), PEOU is closely related to actual behavioral control, which serves as a proxy for actual control in predicting actual behavior (Ajzen, 2024).
ii. **Policy perception mainly influences separation behavior.** First, attitude serves as a mediator in policy perception. Previous studies have shown that PU and PEOU have a significant impact on users' attitudes (Sihombing and Permana, 2023). According to multi-attribute attitude theories, the attributes of an object can affect an individual's attitude toward it (Shin et al., 2023). The primary purpose of the separation policy is to cultivate household waste separation over the long term. PU helps residents perceive

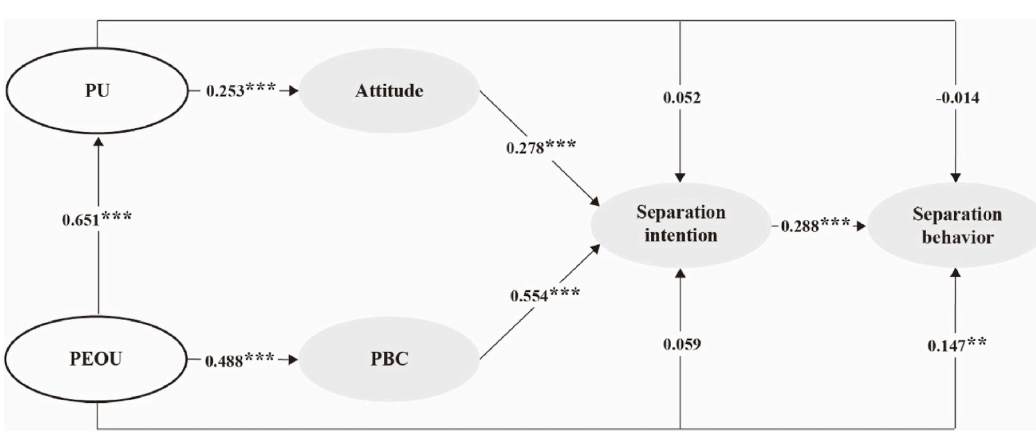

Fig. 4. Hypothesis testing for path analyses. Notes: $^{**}p < 0.05$; $^{***}p < 0.01$.

waste separation as beneficial and effective in solving waste problems, whereas PEOU makes residents aware that waste separation is relatively easy to implement. Therefore, policy perception reflects the attributes of separation behavior and influences residents' attitudes toward it. As the TPB suggests (Ma et al., 2023a), attitude (along with PBC) also influences behavioral intention, which in turn influences behavior. Thus, attitudes mediate the effect of policy perception. Second, PBC mediates the impact of policy perception. PBC includes subjective ability factors (Ajzen, 1991). According to the Self-Efficacy Theory (Poluektova et al., 2023), an individual with a positive PU may overcome separation barriers and enhance their abilities to achieve effective separation. PEOU mainly reflects an individual's perception of the convenience of the separation policy support system, with positive PEOU indicating improved behavioral control and confidence (Sihombing and Permana, 2023). Therefore, policy perception can influence PBC, which in turn mediates the effect of policy perception on PBC.

iii. **The effects of PU and PEOU were different in this study.** As shown in Fig. 5, the total impact of PEOU on the separation behavior was significantly greater than that of PU. If PEOU increases by one unit, separation intention increases by 0.409 units, and separation behavior increases by 0.255 units. However, if PU increases by one unit, the separation intention increases by 0.122 units, and the separation behavior increases by only 0.021 units. In general, waste separation faces several obstacles. Without convenient conditions, such as the provision of knowledge and skills for waste separation and recycling, it is difficult for residents to separate or recycle waste (Yin et al., 2022). Thus, in interviews, many residents pointed out that the inability to determine waste categories is one of the main reasons for their unwillingness to separate waste. PEOU reflects perceived convenience. Therefore, the effect of PEOU on the separation behavior was significant.

iv. **PEOU significantly influenced the PU.** In the path analysis (see Fig. 4), PEOU had a significant effect on PU. This finding is consistent with Alyoussef's (2023) findings. Even if residents have a positive PU, they may abandon compliance with the separation policy if the efforts required exceed the benefits. Hence, PEOU indicates the extent to which a particular behavior requires little effort (Pierce, 2014) and has a greater behavioral impact than PU. Moreover, the formation and strengthening of PEOU depend on the investment of policy resources. PEOU is the objective perception of residents toward the implementation of the separation policy, which mainly follows the cognitive approach of the peripheral route and plays a fundamental role in the cognitive system (Evans, 2008). PU is a relatively subjective and delayed form of perceptual awareness that is formed under the influence of PEOU and other cognitive elements. Therefore, a causal relationship appears to exist between these two policy perceptions.

6.2. Comparison with previous studies

Previous research on the behavioral effects of policy perceptions has yielded mixed results. Some studies have reported a significant influence on separation behavior (Negash et al., 2021; Shen et al., 2022), whereas others have found no meaningful impact (Xu et al., 2017). We argue that these inconsistencies stem from methodological limitations, most notably the lack of quasi-experimental designs, inadequate allocation concealment, and poor masking of the outcome assessments. Critically, many studies incorrectly assume universal awareness of waste separation policies, which undermines the validity of the results. In contrast, our study focuses exclusively on residents who are aware of the policy, thereby improving the reliability of the results and demonstrating the significant behavioral impact of policy perceptions.

Moreover, prior research has often treated policy perception as a single, undifferentiated construct (Fu et al., 2020; Liao et al., 2018; Wan et al., 2014; Wang et al., 2021), limiting the insights into its underlying mechanisms. Our study advances the field by disaggregating policy perception into two key dimensions (PU and PEOU), allowing for a more precise understanding of its operational pathways.

The theoretical contributions of this study are noteworthy. While general behavioral theories, such as the TPB, explain broad pro-environmental actions, our model focuses on policy-driven behavior. Similarly, whereas institutional frameworks, such as the Institutional Analysis and Development Framework, emphasize policy design, our approach bridges the gap between policy design and behavioral response by focusing on individual perception.

6.3. Limitations

One limitation of this study was the subjective nature of the intervention. Policy perception, as defined in this study, refers to individuals' self-reported awareness and understanding of the waste-sorting policy, as assessed through oral interviews. While this approach captures real-world variations in policy exposure, it also introduces potential measurement bias and limits the generalizability of these findings. Residents' perceptions may be influenced by personal interpretation, memory, or social desirability, which could affect the accuracy of group classification. Future research could address this limitation by incorporating more objective or multidimensional measures of policy exposure into the study design. For example, researchers could combine self-

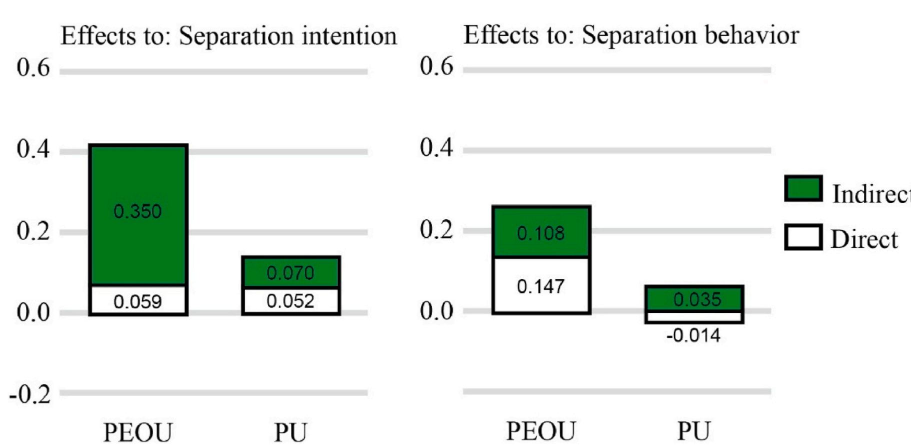

Fig. 5. Standardized total effects of policy perception on separation behavior.

reported perceptions with behavioral indicators (e.g., participation in policy-related events, access to official information channels, or digital trace data) for a more comprehensive perspective.

Furthermore, although we employed sample trimming and statistical tests to ensure demographic comparability between the intervention and control groups, the quasi-experimental design did not allow for randomization of the sample. Therefore, unobserved confounding variables may have existed. Future studies should consider longitudinal designs or instrumental variable approaches to strengthen causal inferences.

7. Conclusions and recommendations

Governments must implement policies to intervene in residents' waste separation behavior, as such behavior is not spontaneous for most people. However, there is limited understanding of the policy's impacts. This study explored this impact, revealing that policy perception significantly influences behavior. Perceived ease of use (PEOU) can directly affect separation behavior, and both PEOU and perceived usefulness (PU) mainly affect behavior through attitude and perceived behavioral control. Additionally, PEOU had a greater impact on behavior and directly affected the PU, which indicates that policy perception is the core explanatory variable linking policies and behavioral changes. This finding partially uncovers the mechanisms linking policy and household waste separation.

Based on the research findings, the following recommendations are proposed to enhance the effectiveness of separation policy interventions.

1. Shifting the government's perspective on environmental policies is crucial. Policy failure may stem not from design flaws but from perception gaps among target populations. Governments should address this issue by incorporating behavioral insights into their policy designs. For example, the UK established the Behavioral Insights Team in 2010 to apply behavioral science to public policy, a strategy that has proven effective and has been replicated in other countries. Similar institutions can help tailor environmental policies to actual behavioral patterns of the population.
2. The scope and depth of residents' policy perceptions should be increased. Governments can raise awareness by intensifying public campaigns, organizing community mobilization events, and diversifying communication channels. For example, Shanghai's "Green Account" program uses digital platforms to track and reward waste separation behavior, thereby enhancing both awareness and engagement.
3. Strengthening residents' PEOU. Given PEOU's significant impact on behavior, the government should improve convenience for residents. This strategy includes providing sufficient waste separation bins, designing clear and easy-to-understand separation signs, and placing bins in easily accessible locations to facilitate waste separation. Japan's success in waste separation can be partly attributed to its user-friendly color-coded bin system and detailed community guidelines.
4. Enhancing residents' perception of the usefulness of the separation policy. Residents are more likely to engage in separation when they observe tangible benefits. Governments should highlight the environmental and social outcomes of separation efforts, as seen in Germany's reports on recycling rates and environmental impact.
5. Design information intervention content based on the characteristics of policy perception. Tailored information campaigns should address both PEOU and PU. For instance, including practical tips and success stories in outreach materials can improve PEOU, whereas highlighting long-term environmental benefits can strengthen PU.

This study provides preliminary evidence for the proposed relationships. However, given the tory nature of the study, further research with representative samples and robust statistical analysis is necessary to validate and extend these findings.

CRediT authorship contribution statement

Zhaoyun Yin: Writing – original draft, Methodology, Formal analysis, Data curation. **Jing Ma:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Canyou Wang:** Writing – review & editing. **Jiajia Cao:** Writing – review & editing. **Haimei Li:** Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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